EAGC 2026 Track 1 Technical Report

Executable World Model Agent - Qualification Submission

Version: v0.17.6
Track: Unseen Environment Exploration and Closed-Loop Task Execution
Evidence boundary: LocalSim + MazeSim final evidence; VirtualHome diagnostics only
Generated: 2026-07-01 UTC

Closed-loop
World Model
Maze Oracle Validation
Docker Reproducible
Official Adapter Ready

Local validation only. Not official hidden-evaluation score.

Abstract

This submission implements a local Track 1 world-model agent prototype focused on closed-loop exploration, structured world-model construction, task planning, exception recovery, and reproducible evidence generation. The final evidence in this build is centered on LocalSim and MazeSim. LocalSim demonstrates an official-style Track 1 loop with exploration, task reception, execution, and recovery. MazeSim provides topology exploration, blocked-edge recovery, anti-loop stress cases, and oracle-based comparison reports where the reference maze is used only for validation.

VirtualHome continuous tooling is implemented fail-closed. A real Unity/VLM partial live diagnostic is included under optional diagnostics, but no validated real Unity continuous + VLM final evidence is included in `sample_outputs`. Mock VirtualHome replay is included only as CI/diagnostic evidence and is not counted as final Track 1 evidence. No model training, fine-tuning, reinforcement learning, or distillation is used; local Qwen/VLM calls, when enabled, are local inference calls through an OpenAI-compatible endpoint.

Evaluation and Verification Results

Check	Command / Artifact	Result
Python compile	python -m compileall src tests tools	passed
Unit/regression tests	python -m pytest -q	175 passed
Docker smoke	python tools/run_test_suite.py --tier docker-smoke	passed
Source package reproducibility	python tools/check_source_package_repro.py --zip-path dist/source.zip	passed
Bundle checksum	checksums/SHA256SUMS.txt	passed
Source zip cleanliness	No outputs/dist/submission_bundle/venv/pycache/bytecode	passed
Local path leak scan	artifact/source JSON + path scan	passed
LocalSim final sample	sample_outputs/local_sim_track1_demo	passed
Maze stress final sample	sample_outputs/maze_stress	passed
Maze anti-loop final sample	sample_outputs/maze_anti_loop	passed
VirtualHome final sample	sample_outputs/virtualhome_exploration	not included
VirtualHome diagnostics	optional_diagnostics/virtualhome_partial_live + virtualhome_mock_replay	included, not final evidence

Boundary: all reported scores and diagnostics are local validation results. Official ranking must be determined by the EAGC hidden checker under the organizer runtime.

Challenge Alignment

Track 1 evaluates whether a system can explore a partially unseen environment, maintain a queryable world model, plan from that model, execute actions, update the model after new observations, and recover from controlled exceptions. This submission is organized around the same dimensions.

Track 1 dimension	Submission evidence
Task completion	LocalSim closed-loop task run with local heuristic score and phase budgets
World-model quality	world_model.json with rooms, topology, objects, relations, states, plans, exceptions, uncertainty, and evidence
Recovery and replanning	LocalSim controlled exceptions; MazeSim blocked-edge, dead-end, and unreachable-goal stress cases
Execution efficiency	step budgets, path-efficiency metrics, fallback counts, and Docker smoke tests
Robustness and safety	fail-closed adapters, no mock VirtualHome final evidence, source package checks, and path-leak scans

Submission Package Contents

Directory	Role
source/source.zip	Clean source package with flat src/ layout, tests, tools, docs, Dockerfile, and assets
docker/	Dockerfile and Docker run examples
reports/	Technical report PDF/HTML/Markdown and build status
disclosures/	Resource disclosure, reproducibility statement, system limitations, and open-source statement
sample_outputs/local_sim_track1_demo/	Main LocalSim closed-loop Track 1 evidence
sample_outputs/maze_stress/	MazeSim topology exploration with reference comparison
sample_outputs/maze_anti_loop/	Anti-loop, dead-end, blocked-shortcut, and unreachable-goal stress evidence
sample_outputs/visual_evidence_demo/	Supporting image-to-world-model evidence demo
optional_diagnostics/ virtualhome_partial_live/	Real Unity/VLM partial diagnostic, not final evidence
optional_diagnostics/ virtualhome_mock_replay/	Mock VirtualHome replay diagnostic only, not final evidence
checksums/SHA256SUMS.txt	SHA256 checksums for bundle files

System Architecture

The implementation is organized as a flat `src/` layout with environment adapters, perception, world-model update logic, planners, executors, validators, and reproducible harness entrypoints. The shared core follows a closed-loop pattern: observe, extract structured facts, update memory, plan or replan, execute, observe again, and audit the run.

- `env_adapters/`: LocalSim, MazeSim, VisualSequence, VirtualHome, and fail-closed official adapter placeholder.
- `world_model/`: schema, store, update helpers, indexing/canonicalization, action effects, and visibility updates.
- `planner/` and `executor/`: action schema, rule planner, replanner, policy interfaces, action grounding, and simulated execution.
- `validators/`: structural, semantic, LocalSim, MazeSim, VirtualHome, leakage, and final-evidence checks.

World Model Design

The world model is serialized as `world_model.json`. It contains agent state, visited rooms, topology nodes and edges, inferred frontiers, blocked edges, objects, support/container context, relations, states, affordances, uncertainty records, plans, replans, policy traces, and exception evidence. Reference or oracle files are validation-only answer keys and are explicitly marked with `reference_used_for_generation=false` and `reference_used_for_validation=true`.

LocalSim Track 1 Evidence

The LocalSim sample demonstrates a deterministic official-style Track 1 procedure with exploration, task reception and planning, execution, and recovery. It is intended for reproducibility and regression checking rather than official scoring.

Metric	Value
Success	True
Score	100.0 / 100
Total Steps	17
Exploration Steps	8
Planning Steps	1
Execution Steps	3
Recovery Steps	5

MazeSim Topology Evidence

MazeSim provides synthetic topology stress tests without external simulator assets. It validates exploration, topology construction, blocked-edge recovery, dead-end handling, and anti-loop behavior. Each output includes a predicted `world_model.json`, agent `episode_log.jsonl`, validation-only `reference_maze.json`, and `comparison_report.json`.

Maze Stress Summary

Metric	Value
Goal reached	True
Steps taken	28
Shortest path length	14
Map coverage	0.88
Topology edge precision	1.0
Topology edge recall	0.926
Reference used for generation	False

Maze Anti-Loop Summary

Metric	Value
Scenarios run	4
Scenarios passed	4
Average edge precision	1.0
Average edge recall	0.979
Total replans	13
Total backtracks	11
Total oscillation count	0

The dead-end comb case intentionally causes repeated dead-end discovery and backtracking. The unreachable-goal case is expected to terminate gracefully with `expected_outcome_met=true` rather than loop indefinitely.

Visual Evidence and VirtualHome Boundary

The bundle contains a small `visual_evidence_demo` as supporting evidence for image-to-world-model extraction and symbolic visual task answering. VirtualHome is treated carefully because a full VirtualHome scene graph can act as simulator ground truth. Final VirtualHome evidence is accepted only when it is a continuous episode with real captured images, real `vlm_frame_extraction`, observation-side action grounding, and a separate reference answer key used only for validation.

VirtualHome category	Status
Final VirtualHome sample in sample_outputs	Not included
Reason	No validated real Unity continuous + VLM final run in this build
Partial live diagnostic	Included under optional_diagnostics/virtualhome_partial_live
Partial live status	partial
Partial live frames / Qwen calls	7 real frames / 7 real vision calls
Partial live reason	insufficient executable action grounding
Mock replay fixture	Included only under optional_diagnostics/virtualhome_mock_replay
Mock evidence level	mock_ci_smoke
Mock prediction mode	mock_visual_extraction
Mock treated as final evidence	No

The partial live diagnostic demonstrates that real Unity frame capture and real Qwen vision calls can run, but action grounding was insufficient. It is therefore correctly classified as optional diagnostics, not final Track 1 evidence.

Official Runtime Adapter Preparedness

An official runtime adapter boundary is included so future hidden-evaluation integration should affect only the adapter/action-translation layer. It fails closed until official runtime details are released and does not silently fall back to LocalSim.

Item	Status
OfficialAdapter included	True
Status	fail-closed placeholder
Silent fallback to LocalSim	False
Integration boundary	future official runtime should implement observe / step / action_schema only

Artifact Integrity and Reproducibility

The submitted Dockerfile builds a lightweight Python agent container. Model weights are not packaged inside the image; local Qwen/vLLM inference is configured through environment variables when needed.

```
python -m compileall src tests tools
python -m pytest -q
python tools/package_source.py
python tools/check_source_package_repro.py --zip-path dist/source.zip
python tools/create_submission_bundle.py
python tools/run_test_suite.py --tier docker-smoke --timeout-seconds 300
```

The source package check excludes generated outputs, virtual environments, build artifacts, `__pycache__`, bytecode files, local simulator caches, and the submission bundle itself.

Compute Budget and Local Test Hardware

Component	Local setup
CPU	AMD Ryzen 9 9950X3D, 16 cores / 32 threads
RAM	128 GB system memory
GPU	NVIDIA GeForce RTX 5090, 32 GB VRAM
OS	Windows 11 with WSL2 Ubuntu 24.04
Containerization	Docker / Docker Desktop for local smoke tests
Model endpoint	Local OpenAI-compatible vLLM/Qwen endpoint when real model calls are enabled

No training, fine-tuning, reinforcement learning, or distillation was performed. The GPU was used for local VLM/Qwen inference experiments and smoke testing only.

Resource Disclosure Summary

- Public resources are used as design references only.
- LocalSim and MazeSim are self-built local development environments.
- MazeSim reference specs are validation-only oracles, not agent inputs.
- VirtualHome scene graph is allowed only as a local reference answer key when real VirtualHome evidence is generated.
- Qwen model weights are not redistributed.
- No online model APIs are required for official-style local execution.
- The Docker image excludes model weights and external simulator assets.

Failure Analysis and Known Limitations

- LocalSim and MazeSim are local development environments, not official hidden evaluation.
- MazeSim stresses topology and replanning but does not replace household manipulation simulators.
- VirtualHome final evidence is not included because no validated real continuous Unity + VLM final run is available.
- VirtualHome partial live reached real frame capture and real VLM calls, but stopped as partial due to insufficient executable action grounding.
- Visual evidence demo is supportive and does not represent physical robot interaction.
- Real AI2-THOR, ProcTHOR, Habitat, and official runtime integration remain future work.
- Official container execution constraints and hidden checker interfaces must be aligned once the organizer runtime is released.

Final Submission Boundary Statement

All scores and metrics in this report are local validation or diagnostic results. They are not official EAGC hidden-evaluation scores. Official ranking must be determined by the EAGC hidden checker under the organizer runtime.

Generated from the current submission bundle. Styling follows the uploaded blue report/resume template while adapting it for a technical report.